A reader's companion to the cutoff-removal theorem for weakly coupled $P(\varphi)_2$

Alexander Stottmeister
Institut für Theoretische Physik, Leibniz Universität Hannover,
Appelstraße 2, 30167 Hannover, Germany

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Read this first. This companion explains a research record, AI-assisted and not refereed, and proves nothing new; human verification is ongoing. This revised edition of 7 November 2026 describes, with its numbers and pages, `part_i/main.tex` of that date (110 pages), in which Appendix D derives the uniform local moment and the item U_{GJS} (A8(vi)) is retired. Audit item C5, a second human reading of the source page images, is open. Figures are illustrations, not evidence.

Abstract

A reading guide to the record of 7 November 2026 on the spatially cutoff $P(\varphi)_2$ model; it proves nothing new. The record's Theorem 12.2 states that, for $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, the full net of cutoff ground states ω_g (no subnets, no averaging) converges in norm on every local algebra $\mathfrak{A}(O)$, at a super-polynomial rate in the distance to the plateau edge; the limit is locally normal, translation and (for even $\mathscr{P}$) $\mathbb{Z}_2$ invariant, and a ground state of the Glimm–Jaffe dynamics in the spectral sense. It is proved from printed statements, with one Glimm–Jaffe–Spencer theorem imported, used also in a case its printed statement covers and its printed proof does not reach; C5's human readings (items 1–4, 9(c), 10–12) are open. We explain why the question was open, how a uniform gap and moment and an exact light cone replace compactness, and what is not claimed.

How to read. Fifteen minutes: this page, §1 (what is proved, under what), §3 with Fig. 1 (the architecture) and §8 (what is not claimed). Forty-five minutes: everything except Appendix A, which points into the record for every step not explained here. Every claim carries its locator in the record: statement number and printed page.

1 What is proved, and under what

The full net converges, with a rate; the one GJS theorem imported, Theorem 1.1.7, is also used in a case its printed statement covers, beyond its printed proof.

Setting. $\mathcal{H}$ is Q-space over the time-zero free field φ_0 of mass $m_0 > 0$ and covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$. For a bounded open interval O, $\mathfrak{A}(O)$ is the von Neumann algebra generated by the Weyl operators $W(f,0) = e^{i\varphi_0(f)}$ and $W(0,h) = e^{i\pi_0(h)}$, $f, h \in C_c^\infty(O)$ real, with π_0 the time-zero momentum field, and $\mathfrak{A}$ is the norm closure of $\bigcup_O \mathfrak{A}(O)$ [R, §2.1, p. 7]. The interaction sits in the states and in the Glimm–Jaffe dynamics α, a g-independent one-parameter automorphism group of $\mathfrak{A}$ [R, A6(ii), p. 13], not in the local algebras. With $V(h) = \lambda \int h(x) :\!\mathscr{P}(\varphi_0(x))\!:_C dx$ (coupling λ, C-Wick

order), where $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ is the interaction polynomial, set $K(g) = H_0 + V(g)$ (H_0 free), $E(g) = \inf \operatorname{spec} K(g)$ and $H(g) = K(g) - E(g)$; Ω_g is its normalised ground vector, $\omega_g = \langle \Omega_g, \cdot\, \Omega_g \rangle$ the cutoff state, and $\beta_t^g = \operatorname{Ad} e^{itH(g)}$. In [R, Definition 2.1, p. 8], $\mathcal{C}_{\mathrm{E}}$ and $\mathcal{C}_{\mathrm{H}}$ are the measurable $[0,1]$-valued functions of compact support on $\mathbb{R}^2$ and on $\mathbb{R}$, and $\mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$ is the plateau class of $g \in C_c^\infty(\mathbb{R})$ with $0 \le g \le 1$ and $g \equiv 1$ on a nonempty open interval, directed by inclusion of the plateau interiors $\{g = 1\}^\circ$; O_r is the open r-neighbourhood of O, and $d(g, O) = \operatorname{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ)$. Throughout, $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}} := \lambda_*/M_*^2$, with M_*, λ_* chosen so that the hypotheses printed for Theorem 1.1.7 of Glimm–Jaffe–Spencer (GJS) [7] hold at bare mass M_* [R, §2.6, p. 20].

The theorem. Below, τ_a are the space translations and θ is the parity automorphism [R, §2.3, p. 9]; $m_1 > 0$ is the GJS decay constant [R, A8(iv), p. 15]; M is the moment constant of [R, (M_2), p. 24]; and Ψ_{m_1} is the rate function Ψ_γ of [R, (29), p. 36] at the gap parameter $\gamma = m_1$, a function of the distance d stated in §6.

Theorem 12.2 of the record (pp. 38–39)

Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. Then there is a state ω_∞ on $\mathfrak{A}$ such that:

(1) *(full-net convergence with rate)* for every bounded open interval O, $\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0$, and $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ whenever $g \equiv 1$ on O_d, with Ψ_{m_1} decaying faster than every inverse power;

(2) *(local normality)* $\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every O;

(3) *(ground state)* ω_∞ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense;

(4) *(invariances)* $\omega_\infty \circ \tau_a = \omega_\infty$ for all $a \in \mathbb{R}$; and for even $\mathscr{P}$, $\omega_\infty \circ \theta = \omega_\infty$;

(5) *(spectra and local moment)*

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty) \quad (g \in \mathcal{C}_{\mathrm{H}}), \qquad \dim \ker H(g) = 1,$$

and $\sup_{g \in \mathcal{C}_{\mathrm{H}}} \|V(h)\Omega_g\| \le M$ for real h with $\|h\|_\infty \le 1$ supported in an interval of length 2.

[...] The constant M of item (5), and through it the rate Ψ_{m_1} of item (1), contain the finite, unevaluated constant $S_2(B_{3\times 5})$ of Theorem C.5.

The elided sentences name the one GJS theorem imported, Theorem 1.1.7 with its transverse case A8(iv)-T, and the inputs of Theorems D.3–D.4, retire A8(vi), and list A1–A7 with [R, Proposition 3.2, p. 21]. Items (1)–(4) are statements along $\mathcal{G}$; item (5) concerns the cutoff Hamiltonians $H(g)$, $g \in \mathcal{C}_{\mathrm{H}}$.

The status. Theorem 12.2 is proved from printed statements only, with one GJS theorem imported: Theorem 1.1.7, which is also used for sharp-time observables across strips transverse to the time-zero line, a case its printed statement covers and its printed proof does not reach (listed separately as A8(iv)-T). The uniform local moment is a theorem of the record (Appendix D). The human readings of audit item C5 that this status rests on, items 1–4 and 10–12 with item 9(c), are open [R, title-page notice, pp. 1–2]; see §4.

What is new. Convergence, not the topology. The unaveraged full net $(\omega_g)_{g \in \mathcal{G}}$ has one limit on every local algebra $\mathfrak{A}(O)$, with a rate, and every w^* limit point of the construction in Glimm and Jaffe's paper III [3] (GJ3 below) coincides with ω_∞ on each $\mathfrak{A}(O)$ [R, Corollary 12.3, p. 39]. This makes ω_∞ canonical for the cutoff family. Uniqueness among the locally normal ground states of $(\mathfrak{A}, \alpha)$, in the spectral sense of [R, Proposition 12.1,

p. 37], is not claimed [R, §1.3, p. 5].

CONTEXT, NOT PART OF THE RECORD. In GJ3, the averaged states of [3, (2.4)] lie, on each local algebra, in a norm-compact subset of the local dual, and their limit points are normal [3, Thm. 2.3, p. 211]; in the representation of any w^* limit point, the generator of time translations has spectrum in $[0, \infty)$ [3, Thm. 2.1, p. 211].

Table 1 sorts the statements this companion explains by what their proofs use; Fig. 1 shows where A8(iv)-T enters. Status words have one meaning each: proved, partial, conjectured, numerical, refuted; *proved* means proved from the imports A1–A8, read with the open human readings of C5 beside it [R, Appendix B, pp. 44–45].

status	statements of the record (printed page)
proved; rests on A8(iv)-T, directly or through M	Theorems D.4 (95), 5.1 (24), 11.1 (36), 12.2 (38), 13.2 (40); Corollaries 5.2 (25), 12.3 (39); Lemmas 10.2 (35), 10.3 (35), 10.5 (36); Propositions 10.4 (35), 11.2 (37).
proved; Theorem 1.1.7 across parallel strips only	Theorem 7.8 (30), with Lemmas 7.1–7.4 (26–28), Proposition 7.5 (28), Lemma 7.6 (29), Theorem 7.7 (29) and Theorem D.3 (90).
proved, independent of A8	Propositions 2.7 (19), 8.1 (31); Lemmas 2.8 (20), 4.4 (23), 9.1 (32), 9.2 (33), C.2–C.4 (48–49), D.1 (79); Theorem C.5 (51).
proved as implications under a stated hypothesis	Proposition 8.2 (32) and Lemmas 9.3 (34), 10.1 (34): under the gap hypothesis (G_γ) (30), which for $P(\varphi)_2$ only Theorem 7.8 supplies. Proposition 12.1 (37): for any $(g_n) \subseteq \mathcal{G}$ with $d(g_n, O) \to \infty$ for every bounded interval O and ω_{g_n} converging in norm on every $\mathfrak{A}(O)$.
not claimed; open	Not claimed: uniqueness of ω_∞ among locally normal ground states of $(\mathfrak{A}, \alpha)$ in the spectral sense, a sharp rate, anything outside the weak-coupling region (Theorem 13.2 is conditional on Hypothesis 13.1), a gap for a GNS Hamiltonian of ω_∞, identification with the Euclidean (Wightman) theory of GJS, Lorentz covariance (not discussed; §8). Not done: a proof of A8(iv)-T, a value of the constant of Theorem C.5 (§4). *Open:* the human readings of C5 items 1–4 and 10–12, with item 9(c) (§9); the Part II gate (II.11) (§8.2).

Table 1: Claims ledger, read from the record's proofs. Appendix A classifies every numbered statement.

In the record: title-page notice, pp. 1–2; §1.3, p. 5; §2.1, p. 7; Definition 2.1, p. 8; §2.3, p. 9; A6, p. 13; A8(iv), p. 15; §2.6, p. 20; Propositions 11.2 and 12.1, Theorem 12.2 and Corollary 12.3, pp. 37–39; Appendix B, pp. 44–45.

2 Why it was open

Compactness gave Glimm and Jaffe a vacuum; it did not give convergence.

Glimm and Jaffe constructed the model in the series I–IV [1, 2, 3, 4]. In paper III, as the record recounts it, the cutoff states are averaged over space translations and a w^*-compactness argument produces a limit point; whether the net (ω_g) itself converges "was not proved there" [R, §1.1, p. 4]. The record quotes the Expositions volume: the $g \to 1$ limit of vacuum expectation values "has not been proved" [6, §4]. It also cites Guerra, Rosen and Simon, who pose Euclidean analogues (local L^p bounds on the Euclidean vacuum density, local convergence of Gibbs states) as Problems 1–2 [8, pp. 125–126]; their note added on p. 128 records Newman's solution of the small-coupling $p = 1$ versions, so "the printed problems must not be read as an unqualified current open-problem claim" [R, §14, p. 41].

Table 2 sets the statements of GJ3 (context) beside those of the record. What the record adds is in the rows *limit* and *selection*, and in the rate of the row *local norm*.

property	GJ3 (context)	the record (weak coupling; through M, resting on A8(iv)-T)
states	ω_n: averages of ω_{g_n} over space translations, $g_n = g(\cdot/n)$, $g \equiv 1$ on $[-3,3]$ [3, (2.3)–(2.5), p. 210]	ω_g for every $g \in \mathcal{G}$; no averaging [R, Theorem 12.2, p. 38]
limit	a w^* limit point; a subsequence of (ω_n) converges w^* to it [3, Thm. 2.1, p. 211]	the limit of the full net in each $\mathfrak{A}(O)^*$ [R, Theorem 12.2(1), p. 38]
local norm	on each local algebra, (ω_n) lies in a norm-compact subset of the local dual [3, Thm. 2.3, p. 211]	Cauchy in each $\mathfrak{A}(O)^*$; rate $\Psi_{m_1}(d)$ when $g \equiv 1$ on O_d [R, Theorem 11.1, p. 36]
normality	every limit point is normal on each local algebra [3, Thm. 2.3, p. 211]	$\omega_\infty\vert_{\mathfrak{A}(O)}$ is normal [R, Proposition 11.2, p. 37]
energy	in the representation of a w^* limit point, the generator of time translations has spectrum in $[0,\infty)$ [3, Thm. 2.1, p. 211]	ω_∞ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense of [R, Proposition 12.1, p. 37]
selection	the record's reading: compactness "selects a state but proves neither convergence nor canonicity" [R, §1.4, p. 5]	every w^* limit point of GJ3's states equals ω_∞ on each $\mathfrak{A}(O)$ [R, Corollary 12.3, p. 39]

Table 2: GJ3 and the record, statement by statement. The GJ3 column is context; its last cell quotes the record.

Moments are not enough. Convergence of Schwinger functions at weak coupling follows from the GJS cluster expansion [R, §1.1, p. 4]: along an affine path of Euclidean cutoffs $h \in \mathcal{C}_{\mathrm{E}}$, GJS bound the change of the moments of products of smeared fields by a constant times $e^{-m_1 d/2}$, with d the distance from the test functions to the change of cutoff, eq. (1.1.11) of [7] as quoted in [R, A8(vii′)(a), p. 17]. In the record's words, such convergence "lives in the wrong topology": "there is no soft passage from moments to norms on local von Neumann algebras" [R, §1.4, p. 5].

Context, not part of the record. The difference $V(g') - V(g)$ has infinite norm, so the machinery for gapped lattice systems of [14, 15] "does not start" [R, §1.4, p. 5].

Later work does not supply the norm statement either, as the record reads it [R, §1.1, p. 4]: Euclidean Gibbs uniqueness under the stated hypotheses of [11] and uniqueness of the invariant measure for the Φ^4_2 stochastic-quantization equation above the critical temperature [12] are not, by themselves, theorems about local-norm convergence of this Hamiltonian cutoff net, and the Hamiltonian renormalisation programme of [13] is restricted to finite volume. The record identifies no such theorem "in the references reviewed here"; this is not a priority claim.

In the record: §§1.1 and 1.4, pp. 4–5; A8(vii′)(a), p. 17; Theorem 11.1, Propositions 11.2 and 12.1, Theorem 12.2 and Corollary 12.3, pp. 36–39; §14, p. 41.

3 The architecture

Two cutoff-uniform numbers, the gap m_1 and the moment M, make the net Cauchy.

The table of the record's §1.2 lays the proof out in steps [R, §1.2, p. 4]. Step (i) is the GJS input, the printed clustering theorem CE2 (Theorem 1.1.7, A8(iv)), including its transverse case A8(iv)-T; step (i′) derives cutoff-uniform moment bounds from it in Appendices C and D (§4). The Euclidean layer (the record's §§4–7) derives the uniform local moment (M_2) and the uniform gap (G) by Feynman–Kac–Nelson (FKN) formulas on slabs, steps (ii)–(iii); from these two the Hamiltonian layer (its §§8–11) derives a local-perturbations-perturb-locally (LPPL) estimate that makes the net Cauchy, with a rate,

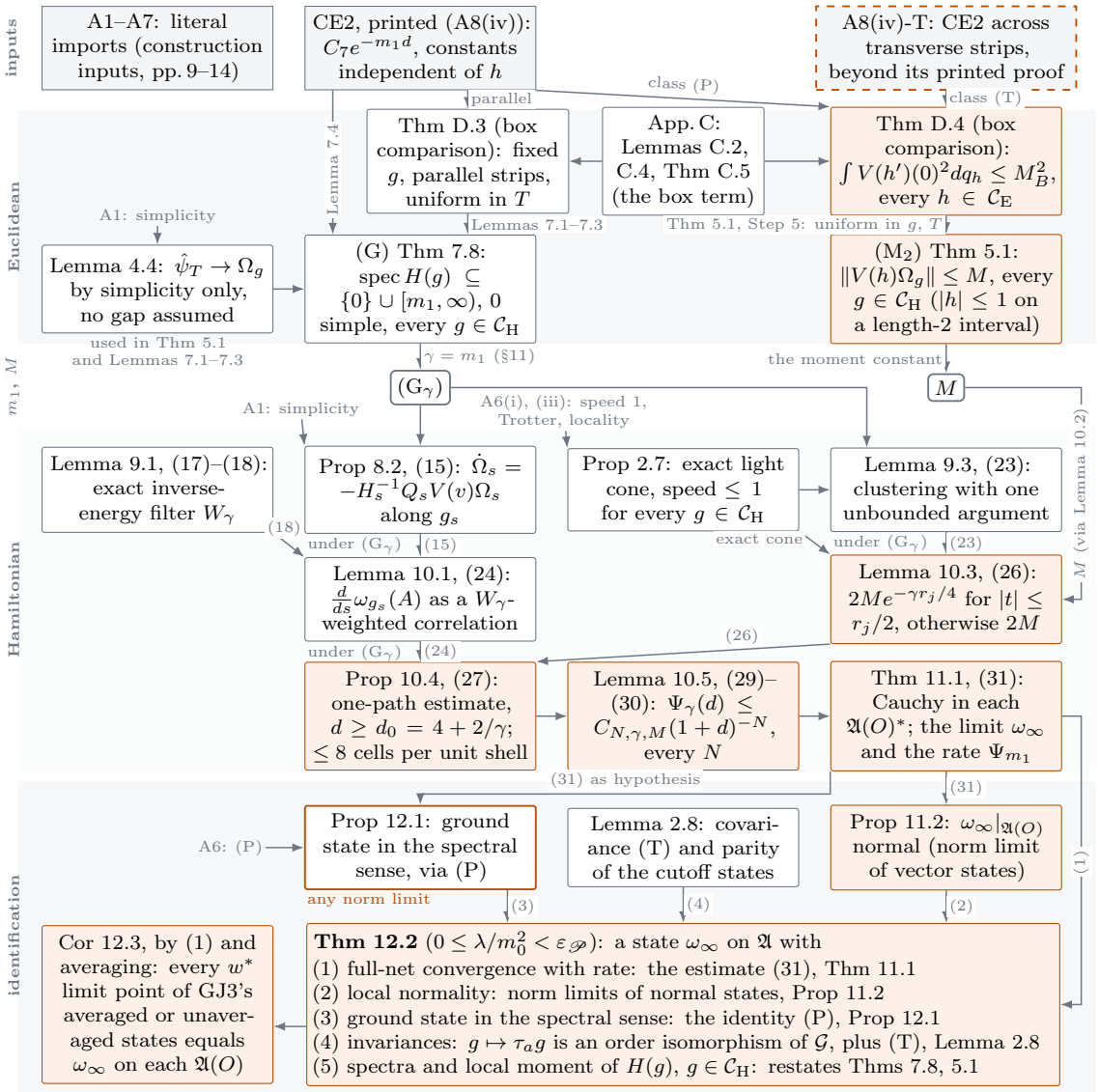


Figure 1: The two layers of the proof and what crosses between them. Boxes name the statements of the record (its Euclidean layer, §§4–7 with Appendices C–D; the interface; its Hamiltonian layer, §§8–11; the identification, §12); arrows say what each proof consumes. Orange marks what rests on A8(iv)-T, the transverse case of GJS Theorem 1.1.7, read from the record's proofs: orange fill, the proof uses it, directly (Theorem D.4) or through the constant M (Theorem 5.1 and every statement that consumes M); orange outline, an implication whose hypothesis, for $P(\varphi)_2$, only an orange node supplies. Unmarked nodes do not use A8(iv)-T; the dashed orange input is A8(iv)-T itself, and the other light-filled inputs and the two interface nodes carry no class. Drawn from [R, §1.2, p. 4] and the proofs named on each node; nothing is computed. Illustration, not evidence.

step (iv); its §12 shows that the limit is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense (Proposition 12.1), with its invariances, step (v). The convergence argument is Hamiltonian; its two inputs come from Euclidean estimates through slab FKN transfer.

Fig. 1 marks in orange what rests on A8(iv)-T, read from the record's proofs: filled, a proof that uses it, directly (Theorem D.4) or through the constant M (Theorem 5.1 and every statement that consumes M); outlined, Proposition 12.1, whose hypothesis (a sequence (g_n) in $\mathcal{G}$ with $d(g_n, O) \to \infty$ for every bounded interval O and ω_{g_n} converging in norm on every local algebra $\mathfrak{A}(O)$) only a filled node supplies for $P(\varphi)_2$. Proposition 8.2 and Lemmas 9.3 and 10.1 are implications under (G_γ) [R, §8, p. 30]: $\operatorname{spec} H(g) \subseteq \{0\} \cup [\gamma, \infty)$ and $\dim \ker H(g) = 1$ for every $g \in \mathcal{C}_{\mathrm{H}}$, which for $P(\varphi)_2$ only Theorem 7.8 supplies, using Theorem 1.1.7 across parallel strips only.

Theorems 5.1 and 7.8 deliver the two numbers, which Theorem 12.2(5) restates for the cutoff Hamiltonians $H(g)$, $g \in \mathcal{C}_{\mathrm{H}}$: the moment bound $\|V(h)\Omega_g\| \le M$ for real h with $\|h\|_\infty \le 1$ supported in an interval of length 2 [R, Theorem 5.1, (M_2), p. 24], resting on A8(iv)-T through Theorem D.4, and the gap $\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty)$ [R, Theorem 7.8, (G), p. 30].

The record's proofs show where each uniformity comes from. The g-independence of m_1 is CE2's printed h-independence [R, Source Fact 2.2, p. 15], applied across a horizontal strip [R, Lemma 7.4, p. 28]. In the record's §7, Lemmas 7.1–7.3 use Theorem D.3 at one fixed g, uniformly in T [R, Lemmas 7.1–7.3, pp. 26–27]. Across g, Theorem D.4 makes M independent of g, at the price of transverse strips (§4), and Lemma 10.2 uses this at every cutoff g_s of an affine path [R, Lemma 10.2, p. 35].

The record's §§8–10 work with an abstract gap γ under (G_γ), and its §§8–11 use the GJS package only through m_1 and M: Lemma 10.2 calls Theorem 5.1, and γ is instantiated as m_1 in [R, §11, p. 36]. Their output, the one-path estimate [R, Proposition 10.4, eq. (27), p. 35], bounds $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$ whenever $g \equiv g' \equiv 1$ on O_d, $d \ge d_0 := 4 + 2/\gamma$. Plateaus are cofinal and local duals are complete, so the net converges in norm in each $\mathfrak{A}(O)^*$, at a rate Ψ_{m_1} faster than every inverse power (§6):

Checkpoint: Theorem 11.1; through M, it rests on A8(iv)-T

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d) \qquad \text{whenever } g \equiv 1 \text{ on } O_d.$$

[R, Theorem 11.1, eq. (31), p. 36]

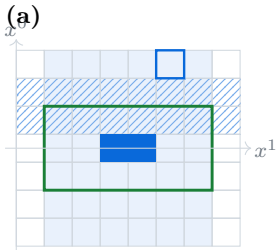
In the record: §1.2, p. 4; Source Fact 2.2, p. 15; Theorem 5.1, p. 24; Lemmas 7.1–7.4 and Theorem 7.8, pp. 26–30; (G_γ), p. 30; Lemma 10.2 and Proposition 10.4, p. 35; Theorem 11.1, p. 36.

4 The uniform local moment, from the printed clustering theorem

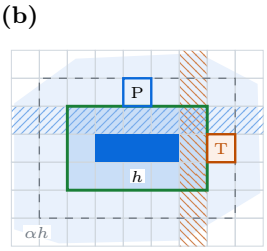
The uniform local moment is a theorem of the record: restrict the cutoff to a box of GJS squares; Theorem 1.1.7 bounds the difference, Appendix C the box.

Of the record's eight imports [R, §2.4, p. 9], A1–A7 quote printed statements (table below). A8 packages the input of Glimm–Jaffe–Spencer [7] for the interacting Euclidean measures $dq_h = Z(h)^{-1} e^{-\mathcal{U}_h} d\mu_0$, $h \in \mathcal{C}_{\mathrm{E}}$, of A3 ($\mathcal{U}_h$ the Euclidean interaction, $d\mu_0$ the free Euclidean measure): the cutoff classes, the observables, the printed norms $\|\cdot\|_{\mathrm{GJS}}$, the dilation, and one theorem, CE2; its items (v)–(vii′) are provenance, used by no proof [R, A8, *Used in*, p. 19].

Theorem 12.2 is proved from printed statements only, with one GJS theorem imported:



Thm D.3: the box B (green) spans every column of supp g; every square outside lies above or below, so the strip (hatched) is parallel



Thm D.4: the box $B_{3\times5}$ and its innermost ring (dashed); a square in the top or bottom row of a ring (P) gets a parallel strip, a side square (T) a transverse one: A8(iv)-T

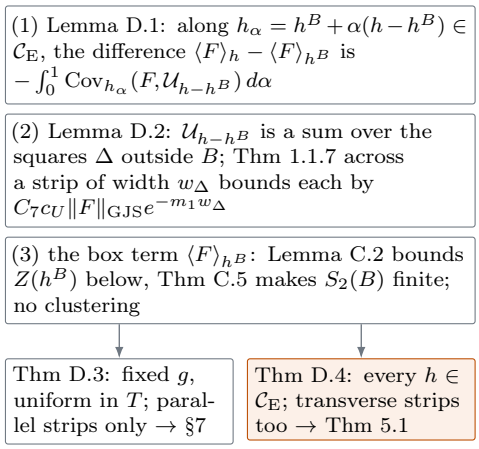


Figure 2: The box comparison of Appendix D, redrawn for this edition (it replaces the figure of 24 October 2026 on two affine paths, which drew the retired reading of A8(vi)). Left: GJS squares (rows in the time x^0, columns in the space x^1), the observable's squares (dark), the box B (green) and charged squares outside it with a separating strip; (a) Theorem D.3, (b) Theorem D.4. The path h_α equals h on B and αh outside it. Right: [R, Lemma D.1, p. 79], [R, Lemma D.2, p. 84], the box term [R, Lemma C.2, p. 48], [R, Theorem C.5, p. 51]. Nothing is computed; no page image is reproduced. Illustration, not evidence.

Theorem 1.1.7, which is also used for sharp-time observables across strips transverse to the time-zero line, a case its printed statement covers and its printed proof does not reach (listed separately as A8(iv)-T). The uniform local moment is a theorem of the record (Appendix D). The human readings of C5 items 1–4 and 10–12, with item 9(c), are open (§9). The imported theorem, as the record transcribes it:

Source Fact 2.2 of the record: CE2 = GJS Theorem 1.1.7 (p. 15)

For localized monomials Q_1, Q_2 with disjoint localization squares separated by a strip of width d bounded by two parallel lines (any orientation),

$$\Big|\int Q_1 Q_2\, dq_h - \int Q_1 dq_h \int Q_2 dq_h\Big| \le C_7 e^{-m_1 d} \|Q_1\|_{\mathrm{GJS}} \|Q_2\|_{\mathrm{GJS}},$$

with $m_1 > 0$ and C_7 *independent of* h, Q_1, Q_2 (so stated in the source).

The comparison. The record's §§5 and 7 need moments of sharp-time observables under the slab measures $dq_{h_{T,g}}$, bounded uniformly in T. Appendix D derives them with the device of GJS's own proof of their (1.1.11) [R, Appendix D, p. 53], in three steps (Fig. 2); $\langle F\rangle_h := \int F\, dq_h$. (1) Restrict $h \in \mathcal{C}_E$ to a box B of GJS squares (unit squares centred at integer points) containing the squares of the observable F, $h^B := h\mathbf{1}_B$. Along $h_\alpha := h^B + \alpha(h - h^B) \in \mathcal{C}_E$, $\alpha \in [0,1]$, the difference $\langle F\rangle_h - \langle F\rangle_{h^B}$ is minus the α-integral of the covariance, under dq_{h_α}, of F with the interaction $\mathcal{U}_{h-h^B}$ outside the box [R, Lemma D.1, p. 79]. (2) That interaction is a sum over the squares Δ outside B charged by h; a strip of positive width w_Δ separates each from the squares of F, and Theorem 1.1.7 bounds its contribution by $C_7 c_U \|F\|_{\mathrm{GJS}} e^{-m_1 w_\Delta}$, with c_U a bound on the GJS norm of the interaction in one square [R, Lemma D.2, p. 84]. (3) The box term needs no clustering: Lemma C.2 bounds $Z(h^B)$ from below, and $\|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)}$ is at most $S_2(B)$, the supremum of that norm over the cutoffs supported in the closed box: finite by Theorem C.5, not evaluated, and dependent on the shape of B only [R, Proposition N.15, p. 62]. Two choices of the box give two theorems, the second resting on A8(iv)-T:

Checkpoint: Theorems D.3 and D.4

Assume the standing assumption of [R, §D.c.1, p. 82]. For $g \in \mathcal{C}_{\mathrm{H}}$, a bounded interval I of positive length, $s \in \mathbb{R}$ and a finite real linear combination F of cylinder monomials supported in I, for every $T > 0$,

$$\left| \int F(s)\, dq_{h_{T,g}} \right| \le A \, \|F\|_{L^2(d\phi_0)} + B' \, \|F(s)\|_{\mathrm{GJS}}, \qquad \text{[R, Theorem D.3, p. 90]}$$

and for every $h \in \mathcal{C}_{\mathrm{E}}$, $a \in \mathbb{R}$ and real measurable h' with $\|h'\|_\infty \le 1$ and $\operatorname{supp} h' \subseteq [a, a+2]$,

$$\int V(h')(0)^2 \, dq_h \le M_B^2. \qquad \text{[R, Theorem D.4, p. 95]}$$

Here $F(s)$ is F at time s, $V(h')(0)$ the local interaction at time 0, $A = e^{3\lambda|a_0|N_B} S_2(B)$, $B' = 2C_7 c_U N_g e^{-m_1}(1 - e^{-m_1})^{-1}$, with $N_g \le N_B$ the column counts of $\operatorname{supp} g$ and of the box; M_B is in §5.

For Theorem D.3 the box is three rows of squares around the time s, across every column from the leftmost to the rightmost one meeting $\operatorname{supp} g \cup I$; every charged square outside lies above or below, so every strip is parallel to the time-zero line [R, Remark N.39, p. 92]. The constants then depend on g, which is all the record's §7 needs: its limits $T \to \infty$ are taken at fixed g. For Theorem D.4 the observable lies in three adjacent squares of row 0, and $B_{3\times 5}$ adds a row above and below and a column on each side. The squares outside lie in rings, the j-th of $8j + 12$ squares: its top and bottom rows, class (P), are separated from the observable by a parallel strip of width j; its side columns, class (T), by a transverse strip of width j, and in rows $-1, 0, 1$ by no parallel strip of positive width at all. With parallel strips alone the sum would grow with the number of columns that h charges, which is not bounded uniformly in $h \in \mathcal{C}_{\mathrm{E}}$; the transverse strips make M_B one number for every h [R, §D.e.4, pp. 96–97].

A8(iv)-T. The record discloses: "GJS Theorem 1.1.7 is printed for localized monomials separated by a strip of any orientation (p. 594). Its printed proof rotates the strip into the time direction (p. 597) and so reaches sharp-time monomials only across strips parallel to the time-zero line. Lemma 7.4 uses only that case; Appendix D also uses the transverse case, listed separately as A8(iv)-T" [R, A8(iv-T), p. 15]. In the record's uses the second monomial is integrated over time and the strip is bounded by two lines of constant x^1. The case enters Theorem D.4, hence Theorem 5.1 and every statement that consumes M, from Lemma 10.2 on; it does not enter the record's §7, whose Theorem D.3 uses parallel strips only. It is listed for transparency, not as another kind of import: the printed statement is used at its printed scope, as every item of A8 is. It is not a narrower form of the retired A8(vi), a moment bound on the slab class that stated no clustering; neither statement is known to imply the other [R, A8(iv-T), p. 15]. The record does not prove it [R, Appendix D, p. 53]; closing it is Phase C of `ugjs/PLAN.md`, optional and not scheduled (`docs/imports.md`). Nor does the record re-derive the cluster expansion behind Theorem 1.1.7, whose printed proof takes its constant from Theorem 1.1.8 at the same h (p. 597) [R, A8, Scope (c), p. 18].

Appendix C, and the history. Theorem 12.2 now consumes Appendix C: Lemmas C.2 and C.4 and Theorem C.5 (with Lemma C.3) enter Theorems D.3–D.4 [R, Appendix C, p. 45], with inputs the record repaired on 7 November 2026 (§9); its audit of the source mechanism is kept, Proposition C.1 being "deliberately limited to the audited mechanisms" [R, Proposition C.1, p. 47]. From 16 August to 30 October 2026 the record carried the cutoff-uniform moment estimate U_{GJS}, A8(vi), as a named hypothesis of Theorem 12.2; on 31 October it read that item off one printed theorem, GJS Theorem 1.1.11 (a re-sourcing,

not a proof). Since 7 November 2026 A8(vi) is retired: Theorems D.3–D.4 supply what the record's §§5 and 7 need, and no proof uses it [R, A8, Scope (a), p. 18].

The seven literal imports [R, §2.4, p. 9]:

	import: what it supplies	source pinpoint
A1	*cutoff construction:* the ground state Ω_g of $H(g)$; the closure of $K(g)$ "has $E(g) = \inf \operatorname{spec} K(g)$ as a *simple* eigenvalue"; the Wick powers in $V(h)$	[7, p. 588]; [2, Thms. 2.2.1, 2.3.1]; [5]; [9, Thm. 5.3]; [10, pp. 10–12, 15–17, 134–138]
A2	*Feynman–Kac–Nelson transfer:* the slab objects $h_{T,g}$, ψ_s, c_g and the Euclidean $\leftrightarrow$ Hamiltonian passage	[10, Thm. III.6, Cor. V.14, Thm. V.15]; [8, Thms. II.14, II.16]
A3	*finite-volume stability:* $\mathcal{U}_h$, built from Simon's Wick powers, in every L^p, and $e^{-\varrho \mathcal{U}_h}$ integrable ($\varrho > 0$): the measures dq_h are well defined	[10, pp. 135–137; Thm. V.2(a), p. 137; Thm. V.7(a), p. 152]
A4	*free hypercontractivity:* the L^p smoothing of the pairing bound (the record's §8) and the moment comparison on the chaoses (Appendices C, D)	[10, Thms. I.17, I.22, III.17]
A5	*domain smoothing; common core:* $\Omega_g \in D(V(h))$, and a common core for all $K(g_s)$ along an affine path	(i) [6, Ch. 7, Thm. 7.4]; (ii) [10, Thm. V.12], [9, Thm. 5.3], [2]
A6	*free propagation; patching; the dynamics α:* (i) speed one, the Trotter formula; (ii) (P): $\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{\lvert t\rvert})$, $\alpha_t(A) = \beta_t^g(A)$ if $g \equiv 1$ on $O_{\lvert t\rvert}$ ($A \in \mathfrak{A}(O)$); (iii) locality: $\mathfrak{A}(O) \subseteq \mathfrak{A}(O')'$ for disjoint O, O'; the cutoff cone is Proposition 2.7	[1, §III]; [2, §§3.5–3.6]; [6, §4.1 and Ch. 4]; (iii) [2, p. 397]
A7	*covariance data:* translation covariance and parity	[2, p. 398]

In the record: §2.4 with A1–A8, Source Fact 2.2, item (iv-T) and the Scope paragraph, pp. 9–19; Appendix C (Proposition C.1, Lemma C.2, Theorem C.5), pp. 45–51; Appendix D (Lemmas D.1–D.2, Proposition N.15, Remark N.39, Theorems D.3–D.4, §D.e.4), pp. 53–97.

5 Euclidean layer: the moment and the gap

The gap is an output: simplicity of the ground state is the only spectral input.

This layer produces, one cutoff $g \in \mathcal{C}_{\mathrm{H}}$ at a time, the moment M and the gap m_1 that §6 consumes; every limit $T \to \infty$ is taken at fixed g, and no gap is assumed [R, §4, p. 22]. Theorem 5.1 rests on A8(iv)-T through Theorem D.4; the record's §7 uses Theorem 1.1.7 across parallel strips only (Lemma 7.4, Theorem D.3). Fig. 3 draws the slab.

I. Slab transfer. With $\Omega_0 \equiv 1$ the free vacuum, $\psi_s := e^{-sH(g)}\Omega_0$ and $c_g := \langle \Omega_g, \Omega_0 \rangle > 0$, the FKN import A2 turns sharp-time moments of the Euclidean field Φ under the normalised slab measure $dq_{h_{T,g}}$ into matrix elements of $e^{-tH(g)}$ [R, Lemma 4.1, p. 22]. As $T \to \infty$, $\hat{\psi}_T := \psi_T / \|\psi_T\|$ converges to Ω_g: ψ_s is $c_g\Omega_g$ plus $e^{-sH(g)}$ applied to the component of Ω_0 orthogonal to Ω_g, whose spectral measure has no atom at 0 by simplicity (A1).

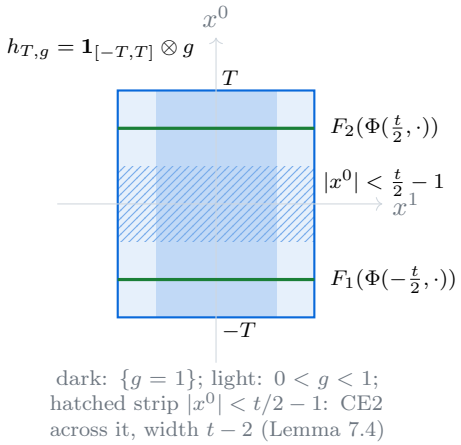
Checkpoint I: transfer at fixed g

For $T > 0$, $t \in [0, 2T]$ and bounded nonnegative measurable F_1, F_2 of the time-zero field,

$$\int F_1\big(\Phi(-\tfrac{t}{2}, \cdot)\big) F_2\big(\Phi(\tfrac{t}{2}, \cdot)\big) dq_{h_{T,g}} = \frac{\langle F_1 \psi_{T-t/2}, e^{-tH(g)} F_2 \psi_{T-t/2} \rangle}{\|\psi_T\|^2}, \qquad \text{[R, (2), p. 22]}$$

and, as $s \to \infty$, "$\hat{\psi}_s \to \Omega_g$ in norm, $\|\psi_s\| \downarrow c_g$"; "*No isolation of the ground-state eigenvalue is used.*" [R, Lemma 4.4, p. 23]

II. The moment. For h as in (M_2) below, Steps 1–4 of the proof of Theorem 5.1 write $Q_V = V(h)^2$ at time 0 as one observable of the GJS class, which admits products of



Euclidean side:
$\int F_1\big(\Phi(-\frac{t}{2},\cdot)\big) F_2\big(\Phi(\frac{t}{2},\cdot)\big)\, dq_{h_{T,g}}$

▼ FKN identity (2), Lemma 4.1

Hamiltonian side: $\dfrac{\langle F_1 \psi_{T-t/2}, e^{-tH(g)} F_2 \psi_{T-t/2}\rangle}{\|\psi_T\|^2}$,
$\psi_s := e^{-sH(g)}\Omega_0$;
all limits at fixed g

▼ Lemma 4.4

$T \to \infty$: $\psi_T/\|\psi_T\| \to \Omega_g$ in norm (Lemma 4.4):
simplicity only, no gap assumed

▼ fixed g: Lemmas 7.1–7.3 with Thm D.3;
CE2 across the strip: Lemma 7.4

Prop 7.5: $\langle \psi_Q, e^{-tH(g)}\psi_Q\rangle \le \widehat{C}_Q\, e^{-m_1 t}$ for centred polynomial vectors ψ_Q;
with density (Thm 7.7): (G)
$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty)$ (Thm 7.8) — an output

Figure 3: Slab transfer, and the gap as an output. Left: the slab cutoff $h_{T,g}$, the two sharp times of [R, Lemma 4.1, eq. (2), p. 22], and the strip across which CE2 is applied [R, Lemma 7.4, p. 28]. Right: the Hamiltonian side of (2), the limit $T \to \infty$ [R, Lemma 4.4, p. 23], and the output, [R, Proposition 7.5, p. 28] and [R, Theorem 7.8, p. 30], with the slab moments of Lemmas 7.1–7.3 bounded by Theorem D.3. Every limit $T \to \infty$ is taken at fixed g, and no gap is assumed: the gap of Theorem 7.8 is the output of this picture. Working coordinates; lengths are not physical. Nothing is computed. Illustration, not evidence.

individually Wick-ordered powers at equal times [R, A8(ii), p. 14]: the support of h meets at most three columns of GJS squares, hence at most nine ordered column pairs, and the printed norm, with constants K_1 and $K_2 \ge 1$ [R, A8(iii), p. 15], bounds $\|Q_V\|_{\mathrm{GJS}}$ [R, Theorem 5.1, eq. (5), p. 24], the bound that Theorem D.4 uses. Step 5 applies eq. (2) at $t = 0$ with $F_1 = F_2 = |V(h)|$ and Theorem D.4 at the slab cutoff $h_{T,g}$, which give $\|V(h)\hat{\psi}_T\| \le M$ for every T [R, eq. (6), p. 25]; the graph of $V(h)$ is weakly closed and $\hat{\psi}_T \to \Omega_g$, so the bound passes to Ω_g:

Checkpoint II: the uniform local moment

Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$ and every real measurable h with $\|h\|_\infty \le 1$ and $\operatorname{supp} h$ contained in an interval of length 2, $\|V(h)\Omega_g\| \le M$ (M$_2$), where

$$M^2 := M_B^2 = e^{15\lambda|a_0|}\, S_2(B_{3\times 5})\, 9^{p'}\, V_* + 9\, C_7\, c_U\, \overline{\Sigma}_{3\times 5}\, \mathfrak{K}(p')\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2,$$

[R, Theorem 5.1, p. 24]

$\mathfrak{K}(p') := K_1 K_2^{8p'} (4p')!\, [\dots]$; M depends on $(\lambda, \mathscr{P}, m_0)$ through C_7, m_1, K_1, K_2, $S_2(B_{3\times 5})$ and V_*, in working coordinates (Remark 3.3), and on nothing else.

Here $S_2(B_{3\times 5})$ is the constant of §4 for a box of 3×5 squares, V_* the supremum of $\|V(h')\|^2_{L^2(d\phi_0)}$ over admissible h', and $\overline{\Sigma}_{3\times 5} = \sum_{j\ge 1}(8j+12)e^{-m_1 j}$ the ring sum [R, §D.e.1, pp. 93–94].

III. Decay at fixed g. Lemmas 7.1–7.3 pass the moments of a cylinder monomial Q of sharp-time fields from the slab measures to Ω_g as $T \to \infty$ [R, Lemmas 7.1–7.3,

pp. 26–27]. Theorem D.3, at the one cutoff g, bounds their square moments uniformly in T: $\sup_{T>0}\|Q\hat{\psi}_T\|^2 \le A\|Q^2\|_{L^2(d\phi_0)} + B'\mathfrak{n}(Q)^2$, with $\mathfrak{n}(Q)^2$ the bound [R, (8), p. 26] on $\|Q^2\|_{\mathrm{GJS}}$ and constants A, B' that depend on g [R, Lemma 7.1, p. 26]. Separately, CE2, with constants C_7 and m_1 [R, Source Fact 2.2, p. 15], applied across the strip $\{|x^0| < t/2-1\}$ of width $t-2$, bounds the connected correlation of $Q(-t/2)$ and $Q(t/2)$ by $C_7 e^{2m_1} e^{-m_1 t}\|Q\|^2_{\mathrm{GJS}}$ for $t > 2$ and every cutoff [R, Lemma 7.4, p. 28]. Together, at fixed g:

Checkpoint III: semigroup decay of centered observables

Let $\psi_Q := Q\Omega_g - \langle\Omega_g, Q\Omega_g\rangle\Omega_g$. Then

$$\langle\psi_Q, e^{-tH(g)}\psi_Q\rangle \le \widehat{C}_Q\, e^{-m_1 t} \quad (t \ge 0), \qquad \widehat{C}_Q := \max\{C_7 e^{2m_1}\|Q\|^2_{\mathrm{GJS}},\ e^{3m_1}\|\psi_Q\|^2\}.$$

[R, Proposition 7.5, p. 28]

IV. The gap. Decay on centred vectors controls the spectrum only if, with Ω_g, they span a dense subspace. They do [R, Theorem 7.7, p. 29], by exponential moments of the sharp-time fields in the vector state of Ω_g, with a constant that depends on g [R, Lemma 7.6, p. 29], which use powers of every total degree as the source's class admits [R, A8(ii), p. 14], strip-analytic characteristic functionals, Fourier uniqueness and martingale convergence. For every $0 \le \beta < m_1$, Checkpoint III forces $\mathbf{1}_{[0,\beta]}(H(g))$ to annihilate every ψ_Q, so by density it is the projection onto $\mathbb{C}\Omega_g$; that is, $H(g)$ has no spectrum in $(0, m_1)$ [R, Theorem 7.8, p. 30]:

Checkpoint IV: the uniform gap

Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$,

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty), \qquad \text{[R, (G), p. 30]}$$

and 0 is a simple eigenvalue.

No gap entered Pieces I–III; (G) is their output. The slab limits use weak compactness in $\mathcal{H}$, and each limit is identified by a closed graph, so nothing is selected.
In the record: §4 with Lemmas 4.1 and 4.4 and eq. (2), pp. 22–23; A8(ii)–(iii) and Source Fact 2.2, pp. 14–15; Theorem 5.1 with eqs. (5), (6) and (M_2), pp. 24–25; Lemmas 7.1–7.4 and 7.6, Proposition 7.5, Theorems 7.7–7.8 and (G), pp. 26–30; Theorems D.3–D.4, pp. 90 and 95.

6 Hamiltonian layer: local perturbations perturb locally

An exact filter turns the gap into decay in time; an exact light cone turns time into distance.

For $g, g' \in \mathcal{G}$, every $\gamma > 0$ satisfying (G_γ) [R, §8, p. 30], and M the moment constant of (M_2), the record's §§8–10 prove the following contract, with Ψ_γ the rate function of [R, (29), p. 36], assembled at the end of this section; through M it rests on A8(iv)-T. Fig. 4 draws the three moves that prove it.

The contract, Lemma 10.5

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le \Psi_\gamma(d) \quad \text{whenever } g \equiv g' \equiv 1 \text{ on } O_d,\ d \ge 0,$$

[R, Lemma 10.5, p. 36]

and for every $N \ge 1$ there is $C_{N,\gamma,M} < \infty$ with

$$\Psi_\gamma(d) \le C_{N,\gamma,M}\,(1+d)^{-N} \qquad (d \ge 0). \qquad \text{[R, (30), p. 36]}$$

V. The derivative along a path. Put $g_s := (1-s)g + sg'$ and $v := g' - g$, $s \in [0,1]$. If

(a)

$$\frac{d}{ds}\,\omega_{g_s}(A) \;=\; -2\,\mathrm{Re}\sum_j \int_{\mathbb{R}} W_\gamma(t)\, C_{j,s}(t)\, dt,$$
$$C_{j,s}(t) \;=\; \langle V_j\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle \;-\; \langle V_j\Omega_s, \Omega_s\rangle\, \omega_{g_s}(A)$$

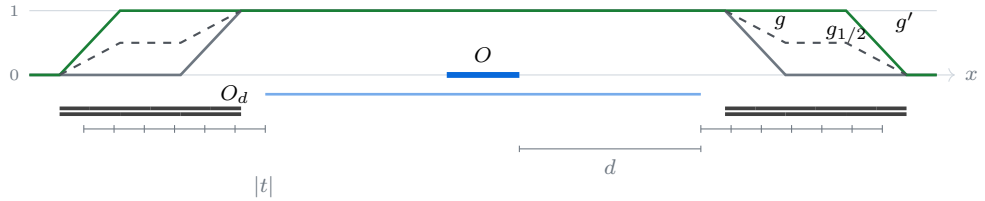
(1) exact filter, Lemma 9.1
$H_s^{-1}Q_s = \int W_\gamma(t)\, e^{itH_s} Q_s\, dt$;
$W_\gamma \in L^1$ with finite moments of every order.

(2) exact light cone, Prop. 2.7
$\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$: speed ≤ 1 for every $g \in \mathcal{C}_{\mathrm{H}}$; no Lieb–Robinson bound is needed.

(3) clustering, Lemmas 9.3, 10.3
$|C_{j,s}(t)| \le 2Me^{-\gamma r_j/4}$ for $|t| \le r_j/2$;
$\|V_j\Omega_s\| \le M$ is the moment (M_2).

(b) $V(v)$, $v = g' - g$: unbounded, not small; only $V_j\Omega_s$ enters (Lemma 10.2). plateau of $g_s \supseteq \{g = 1\}^\circ \cap \{g' = 1\}^\circ$ (Lemma 3.1(iii)).

bars: $B_j = \mathrm{supp}(\eta_j v)$, $j = \pm 8, \dots, \pm 14$ (Lemma 10.2); ticks: the unit shells $r_j \in [n, n+1)$, $n = d, \dots, d+6$; ≤ 4 cells per shell per side (Prop 10.4, Step 3)



(c)

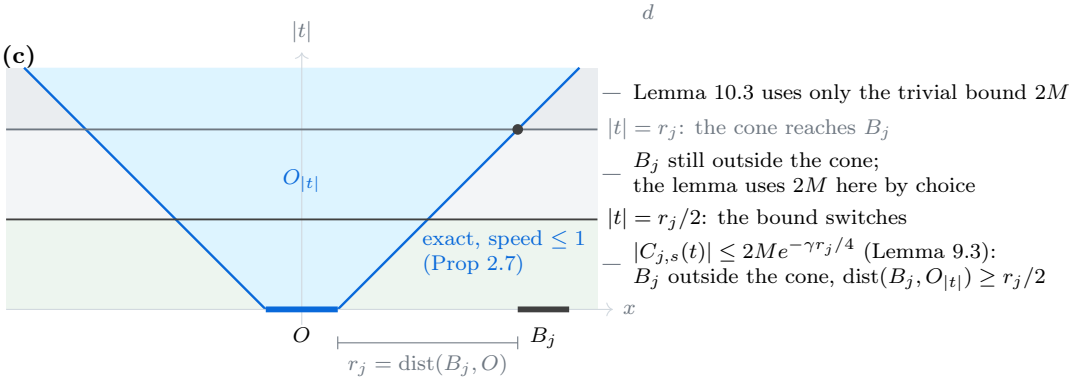


Figure 4: The derivative of the state along the path. (a) The three moves that control it, [R, Lemma 10.1, eq. (24), p. 34] and [R, Proposition 10.4, Step 2, p. 35]. (b) The path g_s and the cells B_j of [R, Lemma 10.2, p. 35], drawn only where $v = g' - g \neq 0$, with the unit shells of [R, Proposition 10.4, Step 3, p. 36]. (c) The two-regime bound of [R, Lemma 10.3, p. 35], drawn to scale for one cell, with the exact light cone of [R, Proposition 2.7, p. 19]. The switch at $|t| = r_j/2$ is a switch of the bound, not of the correlation; the cone reaches B_j only at $|t| = r_j$. No automorphism or spectral flow is constructed: the derivative of the state is bounded and integrated along the path. Nothing is computed. Illustration, not evidence.

$g \equiv g' \equiv 1$ on O_d, every g_s lies in $\mathcal{G}$ and equals 1 on O_d [R, Lemma 3.1(iii), p. 20]; s is a path parameter, not a time. Proposition 8.2, an implication under (G_γ), differentiates the ground vector $\Omega_s := \Omega_{g_s}$ of $H_s := H(g_s)$ in the parallel gauge: $\dot\Omega_s = -H_s^{-1}Q_sV(v)\Omega_s$, with $Q_s := 1 - |\Omega_s\rangle\langle\Omega_s|$ and $\|H_s^{-1}Q_s\| \le \gamma^{-1}$ [R, Proposition 8.2, eq. (15), p. 32]. The exact inverse-energy filter of [R, Lemma 9.1, p. 32] trades this reduced resolvent for real-time evolution: for an even $\chi \in C^\infty(\mathbb{R}; [0,1])$ with $\chi = 0$ on $[-\gamma/2, \gamma/2]$ and $\chi = 1$ outside $(-\gamma, \gamma)$, let W_γ be the inverse Fourier transform of $\widehat{W}_\gamma(E) := \chi(E)/E$ $(E \neq 0)$, $\widehat{W}_\gamma(0) := 0$ [R, eq. (17), p. 32]. If H is self-adjoint with $\mathrm{spec}\, H \subseteq \{0\} \cup [\gamma, \infty)$, then

$$H^{-1}(1 - P_{\ker H}) = \int_{\mathbb{R}} W_\gamma(t)\, e^{itH}(1 - P_{\ker H})\, dt, \qquad \text{[R, Lemma 9.1(ii), eq. (18), p. 32]}$$

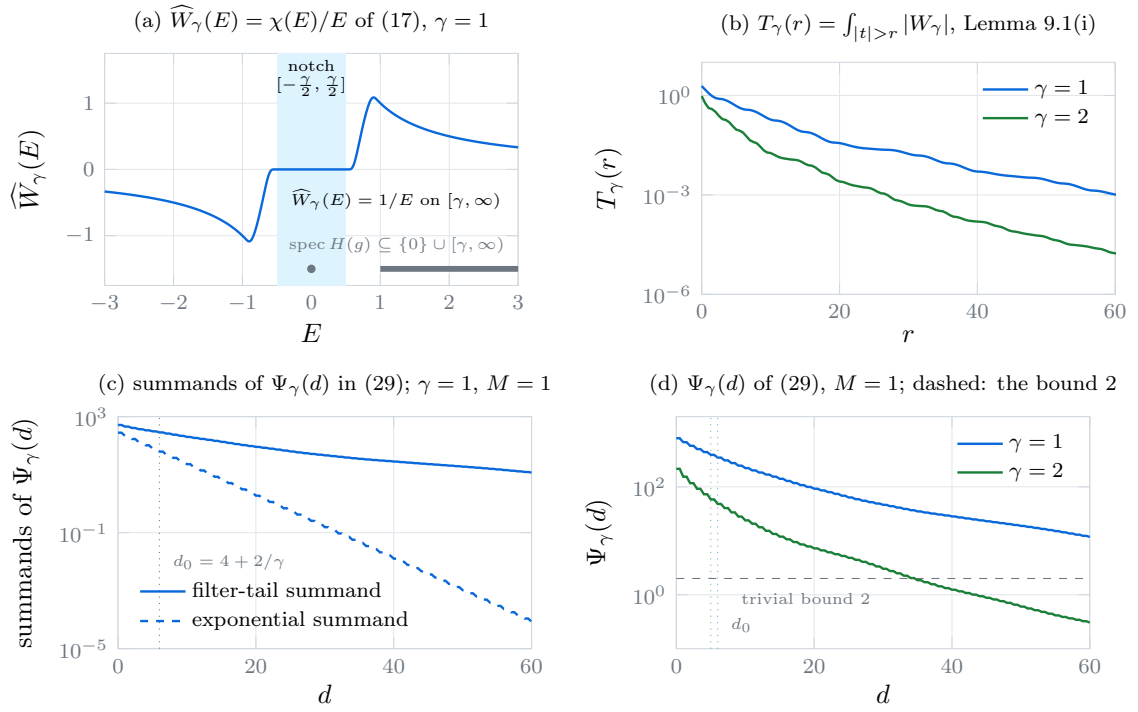


Figure 5: Filter, tail and rate. (a) $\widehat{W}_\gamma(E) = \chi(E)/E$ of [R, Lemma 9.1, eq. (17), p. 32] with its notch; (b) the tail $T_\gamma(r)$ of [R, Lemma 9.1(i), p. 32]; (c) the two summands of $\Psi_\gamma(d)$, [R, Lemma 10.5, eq. (29), p. 36]; (d) $\Psi_\gamma(d)$ with the trivial bound and $d_0 = 4 + 2/\gamma$. Computed by `docs/figures/make_figures.py` for one admissible χ, $\gamma \in \{1, 2\}$ and $M = 1$, on a time grid up to $t = 160$, with values of $|W_\gamma|$ below 10^{-9} set to zero. Lemma 9.1(i) proves only that $W_\gamma \in L^1$ with finite moments of every order, hence $T_\gamma(r) \leq C_{q,\gamma}(1 + r)^{-q}$; the curves are numerical and claim no decay law. Ψ_γ is an upper bound on $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$, never a measured decay of any state; $M = 1$ is a placeholder, not the record's M, which contains unquantified GJS constants and the finite, unevaluated constant of Theorem C.5. At $\gamma = 1$, $\Psi_\gamma(d)$ exceeds the trivial bound 2 at every d shown; at $\gamma = 2$ it falls below 2 from $d = 35$ on: the content is the decay in d, not the size at any d. Both summands decay faster than every inverse power; the exponential summand decays faster, so the filter-tail summand sets the rate. Illustration, not evidence.

an integral identity applied to vectors. Lemma 10.1, an implication under (G_γ), applies it with $H = H_s$ to the single vector $Q_s A \Omega_s$; for self-adjoint $A \in B(\mathcal{H})$ and $0 < s < 1$:

Checkpoint: the state derivative, Lemma 10.1

$$\frac{d}{ds}\,\omega_{g_s}(A) = -2\,\mathrm{Re}\int_{\mathbb{R}} W_\gamma(t)\Big[\langle V(v)\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle - \langle V(v)\Omega_s, \Omega_s\rangle\,\omega_{g_s}(A)\Big]\,dt.$$

[R, (24), p. 34]

VI. From time to distance. Every cutoff dynamics has an exact light cone: $\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ for all $g \in \mathcal{C}_\mathrm{H}$ and $t \in \mathbb{R}$, with speed ≤ 1 independent of g, from A6(i) and the locality of §2.1 (A6(iii)) [R, Proposition 2.7, p. 19]. The perturbation $V(v)$ is unbounded, and (M_2) controls only pieces supported in intervals of length 2, so it is cut into cells $V_j := V(\eta_j v)$, $B_j := \mathrm{supp}(\eta_j v)$, by a smooth partition of unity $\sum_j \eta_j = 1$ with $0 \leq \eta_j \leq 1$ and $\mathrm{supp}\,\eta_j \subset (j-1, j+1)$: then $V(v)\xi = \sum_j V_j \xi$ on common domains and $\|V_j \Omega_s\| \leq M$ for $0 \leq s \leq 1$ [R, Lemma 10.2, eq. (25), p. 35]. Only $V_j\Omega_s$ enters, never $\|V_j\|$. Lemma 9.3, an implication under (G_γ), allows one unbounded argument X: its proof bounds at $t = 0$

two functions of t with spectral supports in $[\gamma,\infty)$ and $(-\infty,-\gamma]$, which the light cone makes agree for $|t|<R$, by the spectral separation [R, Lemma 9.2, eq. (19), p. 33]; the bound needs $\|X\Omega\|$, not $\|X\|$:

Checkpoint: clustering with one unbounded argument, Lemma 9.3

Let $g\in\mathcal{C}_{\mathrm{H}}$, write $H=H(g)$, $\Omega=\Omega_g$, $\beta_t=\beta_t^g$, and assume (G_γ). Let $B\subset\mathbb{R}$ be compact, let $X=X^*$ be affiliated with $\mathfrak{A}(O')'$ for every bounded open interval O' with $\operatorname{dist}(B,O')>0$, with $\Omega\in D(X)$, and let $A\in\mathfrak{A}(O)$ for a bounded interval O. If $R:=\operatorname{dist}(B,O)>0$ satisfies $\gamma R\geq 1$, then

$$\big|\langle X\Omega,A\Omega\rangle-\langle X\Omega,\Omega\rangle\langle\Omega,A\Omega\rangle\big|\leq 2\,\|X\Omega\|\,\|A\|\,e^{-\gamma R/2}. \qquad \text{[R, (23), p. 34]}$$

VII. Two regimes, then shells. Fix $A=A^*\in\mathfrak{A}(O)$ with $\|A\|\leq 1$. For a nonzero cell put $r_j:=\operatorname{dist}(B_j,O)$ and $C_{j,s}(t):=\langle V_j\Omega_s,\beta_t^{g_s}(A)\Omega_s\rangle-\langle V_j\Omega_s,\Omega_s\rangle\,\omega_{g_s}(A)$, the cell's share of the bracket in (24). If $\gamma r_j/2\geq 1$, Lemma 10.3 applies (23) with $X=V_j$ and gives

$$|C_{j,s}(t)|\leq\begin{cases}2Me^{-\gamma r_j/4}, & |t|\leq r_j/2,\\ 2M, & t\in\mathbb{R}.\end{cases} \qquad \text{[R, Lemma 10.3, eq. (26), p. 35]}$$

For $|t|\leq r_j/2$ the cell stays at distance $\geq r_j/2$ from $O_{|t|}$; beyond that, Lemma 10.3 uses only the trivial bound $2M$ (by choice; the cone reaches B_j at $|t|=r_j$). If $d\geq d_0:=4+2/\gamma$ and $g\equiv g'\equiv 1$ on O_d, every cell has $r_j\geq d$, so $\gamma r_j/2\geq 2\gamma+1$ and Lemma 10.3 applies to every cell. Splitting each time integral of (24) at $|t|=r_j/2$ weights the inner bound by $\|W_\gamma\|_1$ and the outer one by the filter tail $T_\gamma(r_j/2)$, where $T_\gamma(r):=\int_{|t|>r}|W_\gamma(t)|\,dt$. Summing the cells shell by shell, at most eight with r_j in any unit shell $[n,n+1)$, four on each side (Fig. 4(b)), and integrating over s gives Proposition 10.4 (Steps 3–4 of its proof):

Checkpoint: the one-path estimate, Proposition 10.4

For $d\geq d_0$ and $g\equiv g'\equiv 1$ on O_d,

$$\|(\omega_g-\omega_{g'})|_{\mathfrak{A}(O)}\|\leq 32M\sum_{n\geq\lfloor d\rfloor}\Big(\|W_\gamma\|_1 e^{-\gamma n/4}+T_\gamma(n/2)\Big). \qquad \text{[R, (27), p. 35]}$$

The rate. Lemma 10.5 names the right side of (27) $\Psi_\gamma(d)$ for $d\geq d_0$, and sets $\Psi_\gamma(d):=\max\{2,\text{the same expression}\}$ for $0\leq d<d_0$ [R, eq. (29), p. 36]. The exponential summand is at most a constant times $e^{-\gamma d/4}$. Lemma 9.1(i) is credited only with $W_\gamma\in L^1$ and finite moments of every order, hence $T_\gamma(r)\leq C_{q,\gamma}(1+r)^{-q}$; this controls the filter-tail summand. Together they give (30), the second display of the contract above. These constants, and Ψ_γ itself, depend on the fixed choice of χ. Both summands decay faster than every inverse power; the exponential summand decays faster, so the filter-tail summand sets the rate. W_γ cannot have exponential tails, because $\widehat{W}_\gamma$ vanishes on $[-\gamma/2,\gamma/2]$ by construction and a transform analytic in a strip cannot vanish on an interval without vanishing identically [R, §1.4, p. 6]. The record states that "Ψ_{m_1} is super-polynomial but not exponential, and this is intrinsic to the filter method used" [R, §1.3(b), p. 5]. Ψ_γ is an upper bound on a norm difference, never a measured decay of any state; Fig. 5 draws its ingredients for one admissible χ.

Context, not part of the record. The machinery of [14, 15] "does not start" here, because $V(g')-V(g)$ has infinite norm (§2; [R, §1.4, p. 5]). No automorphism or spectral flow is constructed: the derivative of the state is bounded and integrated along the path. And no Lieb–Robinson bound is needed: the light cone of Proposition 2.7 is exact.

In the record: §§1.3(b) and 1.4, pp. 5–6; Proposition 2.7, p. 19; Lemma 3.1(iii), p. 20; (G_γ), Proposition 8.2, Lemmas 9.1–9.3 and 10.1–10.3, Proposition 10.4 and Lemma 10.5, with eqs. (15), (17)–(19), (23)–(27) and (29)–(30), pp. 30–36.

7 From Cauchy to the theorem

The ground-state property costs an identity, not an estimate.

Everything in this section rests on A8(iv)-T through Theorem 11.1 and its constant M, except Proposition 12.1 and Lemma 2.8, which do not use A8. Theorem 11.1 turns the one-path estimate of §6 into convergence [R, Theorem 11.1, p. 36]. Step 1: if $\Psi_{m_1}(d) < \varepsilon$, all $g, g' \in \mathcal{G}$ whose plateau interiors contain O_d satisfy $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d) < \varepsilon$ [R, (32), p. 37], so the net is Cauchy in each $\mathfrak{A}(O)^*$. Step 2: completeness of each $\mathfrak{A}(O)^*$ gives limits, compatible under restriction, that define a state ω_∞ on $\mathfrak{A}$. Step 3: the cutoffs equal to 1 on O_d are cofinal in $\mathcal{G}$, so the bound passes to the limit; this is the checkpoint (31) displayed at the end of §3 [R, (31), p. 36].

Local normality comes with the topology: Proposition 11.2 obtains it from the normality of the vector states ω_g on the von Neumann algebra $\mathfrak{A}(O)$ [R, Proposition 11.2, p. 37].

The dynamics α is imported: A6(ii) supplies a one-parameter automorphism group of $\mathfrak{A}$, independent of g, with $\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ and the patching property (P): $\alpha_t(A) = \beta_t^g(A)$ for $A \in \mathfrak{A}(O)$ whenever $g \equiv 1$ on $O_{|t|}$ [R, A6, p. 13]. No continuity of α is imported or used. If $|t| < d(g, O)$, then $g \equiv 1$ on $O_{|t|}$, and (P) with $e^{-itH(g)}\Omega_g = \Omega_g$ gives, for $A, B \in \mathfrak{A}(O)$, the identity of Fig. 6:

$$\omega_g\big(A\alpha_t(B)\big) = \langle \Omega_g, Ae^{itH(g)}Be^{-itH(g)}\Omega_g\rangle = \langle A^*\Omega_g, e^{itH(g)}B\Omega_g\rangle = G_g(t).$$

[R, proof of Proposition 12.1, p. 37]

Here $G_g(t) := \int e^{itE}\, d\rho_g(E)$, where, with $P_{H(g)}$ the spectral measure of $H(g)$, $\rho_g(\cdot) := \langle A^*\Omega_g, P_{H(g)}(\cdot)B\Omega_g\rangle$ is a complex measure with $\operatorname{supp}\rho_g \subseteq \operatorname{spec} H(g) \subseteq [0,\infty)$: inside the window, each two-point function is the Fourier–Stieltjes transform of a measure on $[0,\infty)$. For $|t| \le T$, $A\alpha_t(B) \in \mathfrak{A}(O_T)$ has norm at most $\|A\|\|B\|$, so norm convergence on $\mathfrak{A}(O_T)$ makes these functions converge uniformly on $|t| \le T$, and the definition passes to the limit:

Proposition 12.1 of the record (p. 37)

If $(g_n) \subseteq \mathcal{G}$ satisfies $d(g_n, O) \to \infty$ for every bounded interval O, and $\omega_{g_n} \to \omega$ in norm on every $\mathfrak{A}(O)$, then ω is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense: $\omega \circ \alpha_t = \omega$ for all t; the function $t \mapsto \omega(A\alpha_t(B))$ is continuous and bounded for all $A, B \in \mathfrak{A}$; and

$$\int_{\mathbb{R}} f(t)\,\omega(A\alpha_t(B))\,dt = 0$$

for every $f \in L^1(\mathbb{R})$ whose transform $\check{f}(E) = \int f(t)e^{itE}\,dt$ vanishes on $[0,\infty)$.

Translation invariance needs no averaging: $g \mapsto \tau_a g$ is an order isomorphism of $\mathcal{G}$ with cofinal range, so (T), $\omega_{\tau_a g} = \omega_g \circ \tau_{-a}$ [R, Lemma 2.8, p. 20], and Theorem 12.2(1) give $\omega_\infty \circ \tau_a = \omega_\infty$; parity follows likewise for even $\mathscr{P}$ [R, proof of Theorem 12.2, p. 39].

By Corollary 12.3, every w^* limit point of the averaged or unaveraged cutoff states of [3, §2] coincides with ω_∞ on each $\mathfrak{A}(O)$ [R, Corollary 12.3, p. 39]. Its proof takes GJ3's cutoff function g in $\mathcal{G}$ and applies Theorem 12.2(1), uniformly in the shift, to each shifted cutoff in the average. So ω_∞ is canonical for the cutoff family; its uniqueness is not claimed (§8.1).
In the record: A6, p. 13; Lemma 2.8, p. 20; Theorem 11.1 with eqs. (31)–(32), Propositions 11.2 and 12.1, Theorem 12.2 and Corollary 12.3, pp. 36–39.

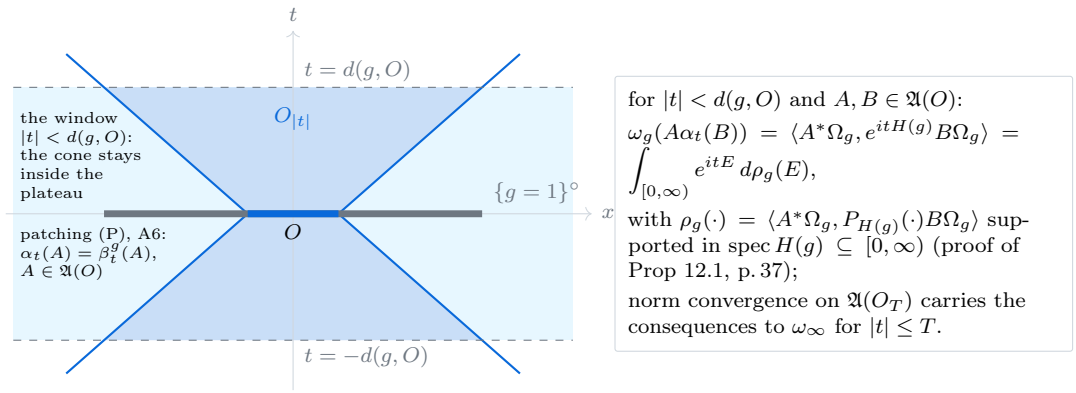


Figure 6: The identity window. The plateau interior $\{g = 1\}^\circ$ of a cutoff g on the axis $t = 0$, a bounded interval O inside it, the cone $O_{|t|}$, and the band $|t| < d(g, O)$ on which the patching identity (P) of [R, A6, p. 13] makes $\alpha_t(A) = \beta_t^g(A)$ for $A \in \mathfrak{A}(O)$. Inside the window the spectral condition is an identity at each cutoff; norm convergence on each $\mathfrak{A}(O_T)$ carries it to ω_∞ (Proposition 12.1). α is Glimm–Jaffe's patched dynamics, imported in A6(ii); the record does not construct it. Drawn from the proof of [R, Proposition 12.1, p. 37]; nothing is computed. Illustration, not evidence.

8 Boundaries, and Part II

Not unique, not sharp, not beyond weak coupling; and Part II is a different question.

8.1 What is not claimed

(a) Uniqueness. Theorem 12.2, which rests on A8(iv)-T through M, identifies the limit of this cutoff family; it does not show that every locally normal ground state of $(\mathfrak{A}, \alpha)$, in the spectral sense of [R, Proposition 12.1, p. 37], equals ω_∞ [R, §1.3(a), p. 5]; [R, §14, *Uniqueness*, p. 41].

(b) Sharp rate. Ψ_{m_1} is super-polynomial but not exponential [R, §1.3(b), p. 5]. A proof of a sharp rate would require "an additional resolvent or Combes–Thomas input not supplied here" [R, §14, *Sharpness*, p. 41]. The record reports, from its working note `m2/` (a proof sketch with numerics, outside the document of record), that in the exactly solvable quadratic model the rate is $\mathrm{poly}(d)\, e^{-2md}$, with m that model's mass.

(c) Beyond weak coupling. No claim is made outside $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ [R, §1.3(c), p. 5]. Theorem 13.2 is conditional on Hypothesis 13.1 [R, Remark 13.3, p. 41]; [R, §14, *Beyond weak coupling*, p. 41], whose four items (B1)–(B4) are all load-bearing [R, §13, p. 39]. For even $\mathscr{P}$ at weak coupling, (B1)–(B4) cannot all hold; the proof uses the uniform gap through Lemma 9.3 and Theorem 12.2 (1), (3), (4), hence also (M_2), and through it rests on A8(iv)-T.

(d) Not addressed. A gap for a GNS Hamiltonian of ω_∞ is not addressed: item (5) concerns the cutoff Hamiltonians $H(g)$, $g \in \mathcal{C}_{\mathrm{H}}$. Nor is ω_∞ identified with the Euclidean (Wightman) theory of GJS; no reverse Osterwalder–Schrader reconstruction is used [R, abstract, p. 2]. Lorentz covariance is not discussed.

8.2 Part II: open at gate (II.11)

Nothing in Part I depends on Part II; §14 of the record states that the two problems have "no mathematical dependence in either direction" [R, §14, p. 41]. Part II asks a different question: uniformity in a lattice or wavelet resolution, in the operator-algebraic renormalisation programme over the wavelet maps of [16] [R, §1.3(d), p. 5], [R, §14, p. 41].

Its gate (II.11) asks that the lattice ground states, evaluated on each fixed coarse Weyl generator, converge as the resolution is refined to the vector state of the ground vector of a continuum torus operator constructed in phase V3 (`part_ii/gate_II11_scoping.pdf`, §1); the scoping pass "proves no part of (II.11)". Already for the free model, no finest-scale local-state-norm error of order $\varepsilon_N^{\vartheta}$, $\vartheta > 0$, is possible [R, §14, p. 41], so the comparison is made at a fixed coarse scale. Status per claim, verbatim from `docs/status.md`:

claim	status
finest-scale diagonal norm convergence	**disproved**
exact finite-torus interacting model and ground state	**constructed**
interacting fixed-coarse projective cluster state along a cofinal subnet	**proved unconditionally**
MMST free fixed-coarse / projective convergence	proved in the source, **strengthened here** to an explicit D4 power rate
exact free gap, group-velocity bound, Wick logarithm	**proved** in the Lamport note
Battle–Federbush as a source for OAR-uniform CE1/CE2	**not available** — the cited note treats different phase-cell variables
full-sequence interacting convergence and identification with the continuum $P(\varphi)_2$ state	**open**, exactly at the coarse Weyl characteristic-functional limit
interacting scale-uniform gap, local second moment, Lieb–Robinson	**open**
the cited Nachtergaele–Raz–Schlein–Sims theorem	**inapplicable** to the Wick quartic: it assumes $V' \in L^1(\mathbb{R})$
the former path $(1-s)H_N + sH$	**undefined** without a common Hilbert-space/form realization; removed from the corrected program

The record is not consistent about this gate: `strategy.md` and the ledger of phase V4 record (II.11) as closed at the stated import scope, `part_ii/README.md` says that phase closure was not yet reached, and the repository README says the gate is open; this companion uses the README's wording.
In the record: abstract, p. 2; §1.3, p. 5; §13 with Hypothesis 13.1, Theorem 13.2 and Remark 13.3, pp. 39–41; §14, p. 41.

9 The record and how to check it

Every import is pinned to a page of its source; nothing here has been refereed.

The record's repository is "an experiment in open science", and the work in it is AI-assisted: the author drafted it with Anthropic's Claude as an interactive assistant and published the working method, including the rounds that found errors in earlier rounds' repairs (`README.md`, "What this is"). Human verification is ongoing.

Gates. The gate table of `docs/status.md` lists the Part I gates, all passed on 10 August 2026: G0 (it compiles), G1/G2 (each step of the two structured proof reconstructions maps to the text), G3a/G3b (constants recomputed by hand), G4 (citations match the dossier), G5 (three `pdflatex` passes, every page inspected), G6 (an adversarial pass, every confirmed finding fixed). The adversarial passes in `strategy.md`, which the README says the assistant ran, are published. Appendix D's drafts and four audit rounds are in `ugjs/`, the gate record of its integration in `part_i/README.md` [R, title-page notice, pp. 1–2]; that record notes that the 110 pages were not each inspected and the dossier index not updated.

Corrections. The corrections history includes audit item P1 (a missing premise, now (B4) of Hypothesis 13.1), S1–S3 (citation pinpoints: FKN transfer, hypercontractivity, domain smoothing), L1–L6 (local repairs, from a cell count to a spectral-parameter range) and gate R2 (the printed-norm prefactor, read from the page image of GJS p. 594) [R,

Appendix B, p. 43]. The scope amendment of 5 September 2026 gave U_{GJS} (A8(vi)) the status of the source's own quantifier, still a named hypothesis; on 31 October 2026 A8(vi) was read off GJS Theorem 1.1.11; each time "No proof in the manuscript changed" [R, title page, p. 1]. On 7 November 2026 the integration of Appendix D retired A8(vi), listed A8(iv)-T, and changed the statements of Theorem 5.1 and Corollary 5.2, which rest on A8(iv)-T, and of Lemmas 7.1 and 7.6 (§4); in the same pass it repaired inputs of Appendix C that were neither proved nor imported, a cell count shifted against the GJS squares, which made the displayed bounds (7)–(8) false as stated (no theorem affected), and the missing construction of the Wick powers in $V(h)$ and $\mathcal{U}_h$ [R, Appendix B, p. 44].

C5. Audit item C5, a check of transcription, not of mathematics, is open: all thirteen verdicts of `part_i/C5_reading_record.md` are blank. Its items 1–4 carry the imports of A8; items 10–11, 12(a),(c) and 9(c) "carry the readings of the Scope paragraph, points (b)–(d), and the item A8(iv)-T"; the rest is corroboration [R, Appendix B, "External residue", pp. 44–45]. Point (d) reads the record's constructed monomials and interaction as GJS's own, which every application of Lemma D.2 uses [R, Lemma D.2, status, p. 84]; so Theorem D.3, behind the gap, names items 1–4 and 10 as open [R, Theorem D.3, status, p. 91].

How to check. Every transcribed display of `part_i/main.tex` carries a `% src:` comment naming its origin, and every import names its source and page images in the audit dossier [R, §2.4, p. 9]; the images are third-party material, not redistributed, and `README.md` names a script that regenerates those of the 16 August 2026 audit. Statement and equation numbers come from `part_i/main.aux`.
In the record: title page, pp. 1–2; §2.4, p. 9; Appendix B, pp. 43–45; Lemma D.2, p. 84; Theorem D.3, pp. 90–91.

Appendix A. Where every statement lives

One row per numbered statement. *Depends on*, read from the proofs: T, uses A8(iv)-T, directly or through a statement of class T (through M or (M_2)); G_γ, assumes (G_γ), nothing of class T; P, uses a printed A8 item (CE2, the observables, the norms, the dilation) and no application of A8(iv)-T; –, none of these; import, A1–A8 and Source Facts 2.2–2.4.

no.	title	page	depends on	treated in, or why routine
2.1	cutoff classes	8	–	§1
A1	cutoff construction	9	import	§§4, 5
A2	Feynman–Kac–Nelson transfer	11	import	§§4, 5
A3	finite-volume stability	11	import	§4
A4	free hypercontractivity	13	import	§4
A5	domain smoothing; common core	13	import	§4
A6	free propagation; patching; the dynamics α	13	import	§§4, 6, 7
A7	covariance data	14	import	§4
A8	the GJS package	14	import	§4
2.2	CE2 = [7, Thm. 1.1.7], p. 594	15	import	§§3, 4, 5
2.3	CE1 = [7, Thm. 1.1.8], p. 595	15	import	provenance; no proof uses it
2.4	[7, p. 629, §4]	17	import	provenance; no proof uses it
2.5	conventions lock	19	–	Wick order equals C-order
2.6	domain smoothing at the ground state	19	–	$\Omega_g \in D(V(h))$ in two lines
2.7	cutoff propagation, uniformly in g	19	–	§6
2.8	covariance and parity of the cutoff states	20	–	§7
3.1	class compatibility	20	–	§6
3.2	dimensionless weak-coupling reduction	21	P	the dilation, A8(viii)
3.3	coordinate convention and transport of the conclusions	21	T	T: transports Theorem 12.2; its coordinate convention, used from [R, §2.6, p. 20] on, involves no A8(iv)-T

no.	title	page	depends on	treated in, or why routine
4.1	slab FKN; bounded nonnegative arguments	22	–	§5
4.2	signed bounded extension	22	–	bounded, then signed
4.3	unbounded extension	23	–	unbounded, by truncation
4.4	slab vacuum; simplicity only	23	–	§5
5.1	uniform local interaction moment (M_2)	24	T	§5
5.2	φ^4 instance	25	T	φ^4 arithmetic; contains S_2
6.1	cell count	25	–	column counting
6.2	GJS norms of polynomial observables	26	P	GJS norms
6.3	row independence	26	P	GJS norms
7.1	moment transfer	26	P	§5
7.2	one-point limit	27	P	§5
7.3	two-point lower semicontinuity	27	P	§5
7.4	CE2 across a horizontal strip	28	P	§§3, 5
7.5	semigroup decay of centered observables	28	P	§5
7.6	exponential moments	29	P	§5
7.7	density of polynomial vectors	29	P	§5
7.8	uniform gap (G)	30	P	§5
8.1	path regularity of the semigroup	31	–	Hölder bookkeeping
8.2	projection and eigenvector regularity; derivative formulas	32	G_γ	§6
9.1	exact inverse-energy filter	32	–	§6
9.2	spectral separation	33	–	§6
9.3	clustering with one unbounded local argument	34	G_γ	§6
10.1	state derivative	34	G_γ	§6
10.2	localization of the perturbation	35	T	§6
10.3	two-regime bound	35	T	§6
10.4	one-path estimate	35	T	§6
10.5	the rate function is super-polynomial	36	T	§6
11.1	full-net convergence with rate	36	T	§7
11.2	local normality	37	T	§7
12.1	exact spectral passage	37	–	§7
12.2	Main Theorem	38	T	§1
12.3	the Glimm–Jaffe vacuum is canonical at weak coupling	39	T	§7
13.1	structure of a broken phase	39	–	§8.1
13.2	conditional exclusion of a two-phase structure	40	T	§8.1
13.3	scope of the phase-transition literature	41	–	§8.1
C.1	consequence of the audited source mechanisms	47	–	the source-mechanism audit
C.2	normalisation	48	–	§4
C.3	Wick lower bound; the $h \geq 0$ entry	49	–	enters Theorem C.5
C.4	ultraviolet difference; the $h \leq 1$ entry	49	–	§4
C.5	h-uniform stability	51	–	§4
C.6	why the class is what it is	52	–	source-mechanism audit
C.7	coverage of every CE application	52	P	lists each use of CE2
D.1	affine-path identity	79	–	§4
D.2	localized comparison	84	T	§4; T: transverse strips
D.3	Phase A: fixed cutoff, parallel strips	90	P	§§4, 5; uses D.2 in the parallel case only (p. 91)
D.4	Phase B: uniform in the cutoff	95	T	§§4, 5

Appendix B. Notation

Symbols with their defining places; Theorems 5.1, 11.1 and what uses M rest on A8(iv)-T.

symbol	meaning
$\mathcal{H}$, m_0	Q-space over the time-zero free field of mass $m_0 > 0$ [R, §2.1, p. 7]
$\mathfrak{A}(O)$, $\mathfrak{A}$	von Neumann algebra generated by $W(f,0)$, $W(0,h)$, $f,h \in C_c^\infty(O)$ real; norm-closed union [R, §2.1, p. 7]
$V(h)$, $\mathscr{P}$	$\lambda \int h(x) :\mathscr{P}(\varphi_0(x)):_C dx$; $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ [R, §2.1, p. 7]
$K(g)$, $E(g)$, $H(g)$	$H_0 + V(g)$; $\inf \operatorname{spec} K(g)$; $K(g) - E(g)$ [R, §2.1, p. 7]
Ω_g, ω_g	normalised ground vector of $H(g)$; its vector state [R, §2.1, p. 7]

symbol	meaning
β_t^g, α	$\mathrm{Ad}\, e^{itH(g)}$ [R, §2.1, p. 7]; patched dynamics with (P) [R, A6, p. 13]
$\mathcal{C}_{\mathrm{E}}$, $\mathcal{C}_{\mathrm{H}}$, $\mathcal{G}$	cutoff classes; $\mathcal{G}$ directed by inclusion of plateau interiors [R, Definition 2.1, p. 8]
O_r, $d(g,O)$	$\{x : \mathrm{dist}(x,O) < r\}$; $\mathrm{dist}(O, \mathbb{R}\setminus\{g=1\}^\circ)$ [R, Definition 2.1, p. 8]
$h_{T,g}$, dq_h	slab cutoff $\mathbf{1}_{[-T,T]} \otimes g$ [R, A2, p. 11]; interacting Euclidean measure [R, A3, p. 11]
$\|\cdot\|_{\mathrm{GJS}}$	printed observable norm (1.1.7)–(1.1.9) of [7] [R, A8(iii), p. 15]
m_1	the GJS decay constant (A8(iv)); also the gap of Thm 7.8 [R, A8(iv), p. 15]
$\varepsilon_{\mathscr{P}}$	λ_*/M_*^2; the region is $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ [R, §2.6, p. 20]
$S_2(B)$	stability constant of a box B; finite, not evaluated [R, Definition N.14, p. 62]
M	constant of (M_2); $M = M_B$ (Thm D.4) contains $S_2(B_{3\times 5})$ [R, Theorem 5.1, p. 24]
Ψ_γ, d_0	rate function, eq. (29) [R, Lemma 10.5, p. 36]; $d_0 = 4 + 2/\gamma$ [R, Proposition 10.4, p. 35]
ω_∞	the limit state [R, Theorem 11.1, p. 36]

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